Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following are best practices associated with Streaming applications?

Options :

6406532335421. ✔ Use a message store that supports message replay so that no data is lost in processing.

6406532335422. ✖ “Hot potato” principle is when the streaming application operates on data from cache (i.e. the hot area of memory) and therefore is able to produce very high throughput

6406532335423. ✖ Hadoop is best suited for executing Streaming applications

6406532335424. ✔ Use checkpointing when faced with mission-critical workloads that require 100% accuracy.

6406532335425. ✖ Handle state pollution by restarting the persistent store software periodically.

Speech Tech

Section Id :	64065349328
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	3
Number of Questions to be attempted :	3
Section Marks :	40
Display Number Panel :	Yes
Section Negative Marks :	0
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes

Maximum Instruction Time : 0

Sub-Section Number : 1

Sub-Section Id : 640653103769

Question Shuffling Allowed : No

Is Section Default? : null

Question Number : 204 Question Id : 640653699411 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY(PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532335518. ✔ YES

6406532335519. ✖ NO

Sub-Section Number : 2

Sub-Section Id : 640653103770

Question Shuffling Allowed : Yes

Is Section Default? : null

Question Number : 205 Question Id : 640653699412 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 8

Question Label : Multiple Choice Question

1.1 MCQ *(Note: Please explain your choice)* 8 points

1. (2 points) Consider a language with a total of 50 phonemes, how many separate models for a single phoneme will need to be created considering a tri-phone model setup?
   A. 2401
   B. 2500
   C. 3
   D. None of these
2. (2 points) Recurrent Neural Networks(RNN) with basic RNN cells exhibit which of the following problems?
   A. Vanishing gradient problem
   B. Exploding gradient problem
   C. both (A) and (B)
   D. None of the above
3. (2 points) What is the primary purpose of an activation function in a neural network?
   A. To normalize the input data
   B. To introduce non-linearity to the network
   C. To control the learning rate
   D. To regularize the weights
4. (2 points) Which of the following is correct according to Bayes rule?
   A. $P(A|B) = \frac{P(A)}{P(B)}$
   B. $P(A|B) = \frac{P(A \cap B)}{P(A)}$
   C. $P(A|B) = \frac{P(A \cap B)}{P(B)}$
   D. $P(A|B) = \frac{P(A \cup B)}{P(B)}$

Options :

6406532335520. ✔ I have written answers on the answer sheets

6406532335521. ✖ Not applicable

Sub-Section Number :	3
Sub-Section Id :	640653103771
Question Shuffling Allowed :	Yes
Is Section Default? :	null

Question Number : 206 Question Id : 640653699413 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 32

Question Label : Multiple Choice Question

1.2 Subjective Questions — 32 points

1. (2 points) Which of the following are valid activation functions you could use to train neural networks? Draw graphs representing the result of using activation function at the input, similar to graphs for activation functions shown in the class. Justify your answer.
   - $f(x) = sin(x)$
   - $f(x) = 0.2x^2 + x + 1$
2. (2 points) Recurrent Neural Networks do not need position encoding. Explain.
3. (3 points) Explain how the HMM and GMM are used in classical ASR.
4. (5 points) Explain speaker embedding based speaker diarisation in detail.
5. (10 points) Explain the FastSpeech TTS system in detail.
6. (10 points) Describe the difference in the attention mechanisms of LSTM-based encoder-decoder model and of the Transformer. Also explain the multi-head self attention in Transformers in detail.

Options :

6406532335522. ✔ I have written answers on the answer sheets

6406532335523. ✖ Not applicable

SPG

Section Id :	64065349329
Section Number :	10
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	23
Number of Questions to be attempted :	23